# Remaining Useful Life Prediction of Rolling Bearings Based on Physics-Informed Neural Networks

**Cheng-Yeh Chiang**
Institute of Statistics
National Yang Ming Chiao Tung University
allenchiang.sc14@nycu.edu.tw

## Abstract

Rolling bearing Remaining Useful Life prediction plays an important role in predictive maintenance and intelligent manufacturing. Traditional model-based prognostic approaches often rely on predefined degradation functions, which may not adequately represent the complex and fluctuating degradation behavior observed in real-world bearing systems. To address this limitation, this study proposes a Physics-Informed Neural Network enhanced Unscented Particle Filter framework for RUL prediction. The proposed method replaces the fixed degradation model in the conventional EM-UPF framework with a neural network and incorporates a physics-informed monotonicity constraint to guide the predicted health trajectory toward physically consistent degradation behavior. Experiments were conducted on public bearing datasets using vibration-based health indicators. The results indicate that the proposed framework produces smoother degradation trajectories and achieves more stable and reliable RUL predictions than the baseline approach. These findings demonstrate the potential of integrating physical knowledge with data-driven models for bearing prognostics and predictive maintenance.

## 1 Background and Motivation

### 1.1 Background

Rolling bearings are essential components in rotating machinery and are widely employed in manufacturing systems, transportation equipment, and various industrial applications. Bearing failures can lead to unexpected downtime, reduced production efficiency, increased maintenance costs, and even serious safety concerns. Therefore, effective monitoring of bearing health conditions and accurate prediction of their Remaining Useful Life are crucial for supporting predictive maintenance and improving system reliability.

### 1.2 Motivation

The motivation of this study comes from the limitation of predefined degradation models in model-based prognostics. In many existing approaches, the degradation process is represented by a fixed mathematical form, such as an exponential or polynomial function. However, real bearing degradation is often nonlinear, non-stationary, and highly fluctuating, so a predefined model may not match the actual degradation trajectory well. This mismatch can reduce the accuracy and stability of remaining useful life prediction. Therefore, this study introduces a physics-informed neural network constraint into the UPF-based prediction framework to guide the estimated health trajectory toward a more physically consistent form and improve RUL prediction performance.

### 1.3 Problem Definition

This study aims to reduce the limitations of conventional degradation prediction models that rely on predefined mathematical assumptions. In addition, it seeks to enhance the stability and reliability of remaining useful life prediction by integrating a Physics-Informed Neural Network into the Unscented Particle Filter framework.

# 2 Literature Review / Related Works

## 2.1 Overview of Related Studies

The main reference of this project is the DUPF study by Cui et al., which proposes a dual unscented particle filter framework for bearing RUL prediction under fluctuating degradation. The method first detects the degradation stage by comparing a healthy reference window with the current health-indicator window using Euclidean distance. After degradation onset is identified, HM-UPF is applied to estimate the hidden degradation state. In the prediction stage, the model combines long-term and short-term degradation information and estimates RUL through a UPF-based sequential framework.

## 2.2 Methodological Comparison

Compared with the original DUPF design, this project adopts a simplified single-stream implementation. The baseline keeps the main sequential prediction logic of the paper, but does not include the full dual-stream fusion mechanism. On this basis, the PINN version further replaces the fixed exponential degradation function with a neural network constrained by monotonicity.

## 2.3 Connection to this study

The literature provides the structural foundation of this project, including TSD detection, HM-UPF state estimation, and sequential RUL prediction. My implementation follows this overall logic, but focuses on a more compact single-stream version and introduces a PINN-based degradation model to improve prediction flexibility under fluctuating degradation.

# 3 Methodology

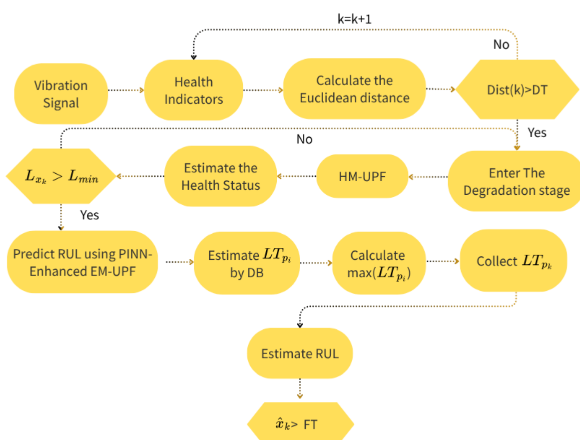


Figure 1: Overall methodology flow of the proposed framework.

Figure 1 summarizes the overall workflow of this study. The process starts from raw vibration signals, which are converted into health indicators. A sliding-window distance is then computed to determine whether the bearing has entered the degradation stage. Once degradation is detected, HM-UPF is used to estimate a smoother latent health state. When the sequence reaches the prediction start point, the model predicts the future degradation evolution and estimates RUL until the state reaches the failure threshold.

Based on this workflow, the methodology can be divided into four main parts: health-indicator construction, degradation-stage detection, hidden-state estimation with failure-threshold setting, and RUL prediction.

**Health Indicator Construction**

Each vibration snapshot is converted into a root mean square value:

$$\mathrm{RMS} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2},$$

where $x_i$ is the vibration amplitude at sample $i$.

**TSD Detection**

A healthy reference window is selected from the early stage. A sliding window of the same length then moves through the health-indicator sequence. The Euclidean distance between the healthy window $X$ and current window $Y$ is computed as

$$\mathrm{Dist}(X, Y) = \sqrt{\sum_{i=1}^{M} (X_i - Y_i)^2}.$$

When the distance exceeds a threshold for several consecutive windows, the bearing is considered to have entered the degradation stage. The corresponding point is defined as the time to start degradation.

**HM-UPF State Estimation**

After TSD is detected, HM-UPF is used to estimate a smoother hidden degradation state from the noisy HI sequence. This step helps reduce the influence of measurement fluctuation and provides a more stable state trajectory for later prediction.

**Failure Threshold and TSP**

The failure threshold is estimated from healthy-stage statistics. In the current implementation, the base threshold is computed from the healthy-state mean and standard deviation:

$$FT_{\text{base}} = \mu_{\text{healthy}} + \lambda\sigma_{\text{healthy}}.$$

To avoid setting the threshold unrealistically close to the prediction start point, the final threshold is defined as

$$FT = \max(FT_{\text{base}}, x_{\text{TSP}} + \Delta),$$

where $x_{\text{TSP}}$ is the estimated state at the time to start prediction, and $\Delta$ is a safety margin. This design keeps the threshold consistent with both healthy-stage statistics and the current degradation level.

**Baseline EM-UPF**

The baseline RUL predictor uses a single-stream EM-UPF. Its role is to propagate the post-TSP degradation state forward and estimate the future failure time. The RUL is then obtained from the difference between the predicted failure time and the current prediction time.

**PINN-UPF**

The PINN version preserves the same TSD, HM-UPF, TSP, and failure-threshold logic, but replaces the fixed exponential degradation function with a neural network. The network takes time as input and predicts the future HI trajectory. To incorporate physical prior knowledge, the training loss contains both a data-fitting term and a monotonicity penalty:

$$\mathcal{L} = \lambda_{\text{data}}\mathcal{L}_{\text{MSE}} + \lambda_{\text{phys}}\mathcal{L}_{\text{mono}}.$$

The monotonicity penalty discourages negative slopes in the post-degradation HI trajectory, so the predicted degradation trend remains more physically consistent and does not repeatedly rebound.

**Method Justification**

These methods are suitable for the problem because they combine an interpretable HI construction process, a clear degradation-stage detection rule, a sequential filtering framework for noisy degradation states, and a PINN-based prediction model that is more flexible than a fixed exponential assumption.

# 4 Data Collection and Analysis Result

## 4.1 Data Collection

This project uses two public run-to-failure bearing datasets: the XJTU bearing dataset and the IMS bearing dataset. The XJTU dataset contains multiple bearings under different operating conditions, and each snapshot provides two-channel vibration measurements. The IMS dataset provides long-term run-to-failure bearing vibration records that are also converted into RMS-based health indicators.

Before modeling, basic data quality checks are conducted, including file-shape consistency, duplicate-snapshot inspection, and invalid-value filtering.

## 4.2 Analysis

The implemented workflow starts from vibration-to-RMS transformation, then applies sliding-window Euclidean distance to detect TSD. After that, HM-UPF estimates the hidden degradation state. TSP and failure threshold are determined from the state sequence, and the RUL is predicted by baseline EM-UPF or PINN-UPF. The following analysis is organized in the same order for both datasets: first the HI curve and TSD detection result, then the baseline and PINN prediction curves, and finally the RMSE and CRA comparison.

### XJTU Case Study

Figure 2 shows the HI and distance curve of the XJTU bearing. The lower panel shows that the sliding-window distance remains near the healthy baseline for a long period and then sharply rises above the threshold. Once the distance persistently exceeds the threshold, the bearing is considered to have entered the degradation stage. In this case, the detected TSD occurs near the end of the life cycle, which is consistent with the rapid growth observed in the RMS HI curve.

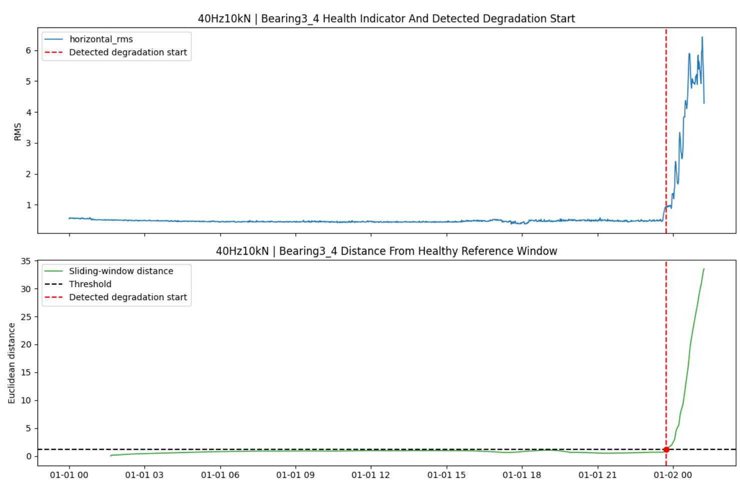


Figure 2: XJTU health indicator and detected degradation start.

Figures 3 and 4 compare the baseline EM-UPF and PINN-UPF RUL predictions. The baseline EM-UPF shows large oscillation and repeated over- and under-estimation, indicating that a fixed exponential trend is not flexible enough for this degradation pattern. By contrast, the PINN-UPF prediction is smoother and stays closer to the actual RUL trajectory over most of the prediction horizon.

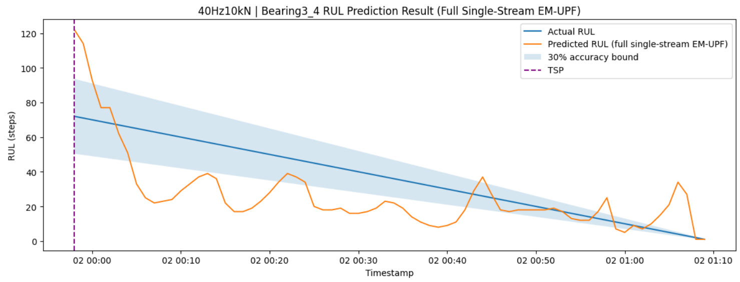


Figure 3: Baseline EM-UPF RUL prediction on the XJTU case.

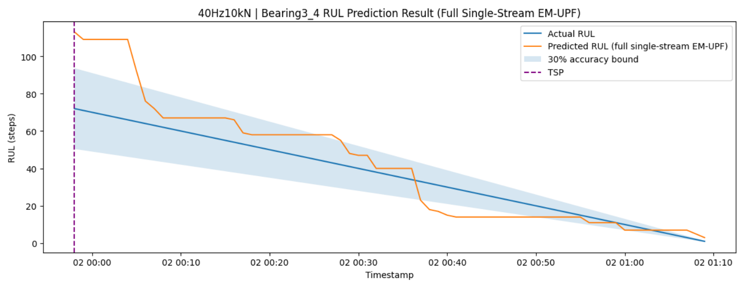


Figure 4: PINN-UPF RUL prediction on the XJTU case.

Table 1: Prediction performance comparison on the XJTU case.

| Method | RMSE | CRA |
|---|---|---|
| Baseline EM-UPF | 21.1187 | 0.5601 |
| PINN-UPF | 15.3333 | 0.7051 |

The XJTU result shows that PINN-UPF reduces RMSE by about 27.4% and increases CRA by about 25.9% compared with the baseline. This indicates that the neural-network-based degradation model can better follow the rapid and irregular late-stage degradation pattern while maintaining a physically plausible monotone trend.

### IMS Case Study

Figure 5 shows the HI and distance curve of the IMS bearing. Similar to the XJTU case, the lower panel indicates that the sliding-window distance first stays close to the healthy baseline and then crosses the threshold at the detected TSD. After this point, the HI starts to show a clearer upward drift and eventually enters a stronger degradation region, which justifies switching from healthy monitoring to degradation-state estimation and RUL prediction.

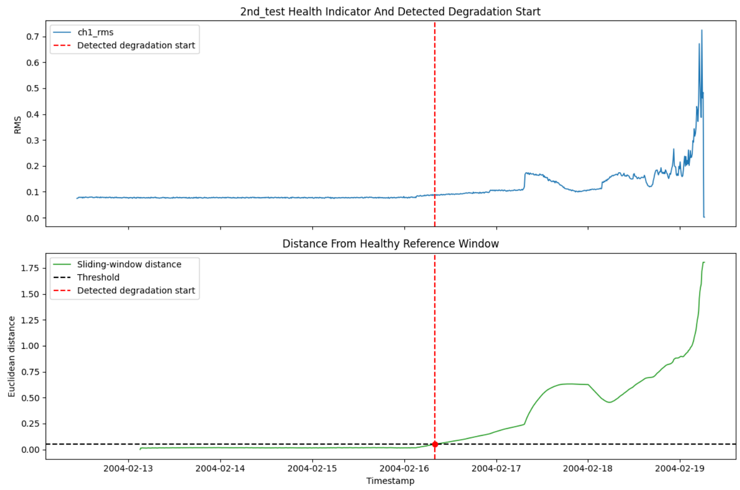


Figure 5: IMS health indicator and detected degradation start.

Figures 6 and 7 compare the baseline EM-UPF and PINN-UPF on the IMS dataset. The baseline EM-UPF remains almost flat over much of the prediction horizon and fails to follow the actual RUL decrease well. In contrast, the PINN-UPF better adapts to the observed degradation evolution and produces a prediction trajectory that more closely tracks the actual RUL line.

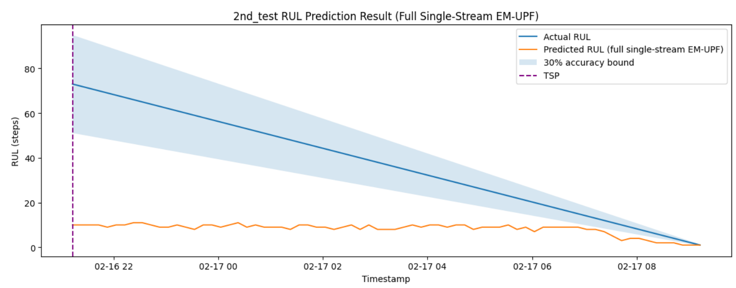


Figure 6: Baseline EM-UPF RUL prediction on the IMS case.

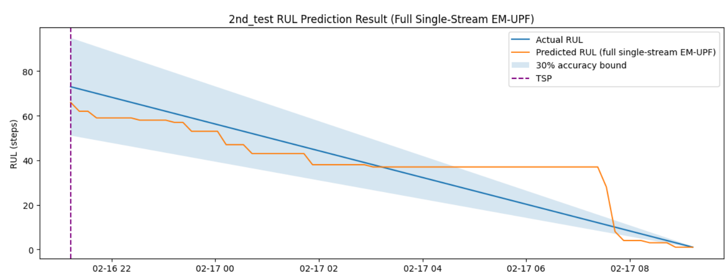


Figure 7: PINN-UPF RUL prediction on the IMS case.

Table 2: Prediction performance comparison on the IMS case.

| Method | RMSE | CRA |
|---|---|---|
| Baseline EM-UPF | 34.6813 | 0.2218 |
| PINN-UPF | 10.0267 | 0.8075 |

The IMS result shows an even larger performance gap. PINN-UPF reduces RMSE by about 71.1% and substantially improves CRA, suggesting that the monotonic neural degradation model is especially helpful when the baseline exponential predictor cannot adapt to the actual degradation evolution.

### 4.3 Interpretation

Across both datasets, the main finding is that the PINN-enhanced model produces smoother and more stable sequential RUL prediction than the baseline EM-UPF. This is reflected not only by lower RMSE, but also by higher CRA, which means the prediction remains more useful over the full prediction horizon instead of matching the target only at isolated time points.

From a maintenance perspective, this is important because RUL is updated continuously in practice. A model with better CRA provides more reliable support for maintenance scheduling and decision making. At the same time, the current study still has limitations. The TSD result depends on the selected threshold and HI definition, and the full XJTU dataset still contains several bearings with highly irregular degradation patterns that remain difficult to model robustly. Therefore, the present results should be interpreted as focused case studies on representative bearings rather than a full robustness validation over every bearing and both channels.

## 5 Conclusion

The experimental results show that the PINN-enhanced model produced smoother and more physically consistent degradation trajectories than the baseline EM-UPF. On both the XJTU and IMS case studies, the PINN-based predictor followed the degradation trend more stably and reduced unrealistic fluctuation in the predicted RUL trajectory.

Compared with the baseline EM-UPF, the proposed method achieved lower RMSE and higher CRA on both datasets. These results suggest that incorporating physics-informed constraints helps improve both prediction accuracy and temporal stability under fluctuating degradation conditions.

For future work, the framework will be extended to include all bearings rather than only representative case studies, so that the evaluation can be more complete across different degradation patterns. In addition, the dual-stream DUPF prediction mechanism in the original literature will be incorporated to compare the current simplified single-stream framework with a more complete fusion-based prediction design.

## References

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